

VISHWAH SIVAGURUNATHAN

King George Avenue, Sarjapura, Bengaluru, Karnataka 562125

+91 7339398959

[Portfolio](#)

vishwahsivagurunathan@gmail.com

[Linkedin](#)

[GitHub](#)

[Shadertoy](#)

Summary

Graphics and game engine programmer with a Master's in Computer Game Engineering (Distinction) from Newcastle University. Proficient in modern C++, HLSL, and Slang with hands-on experience in Vulkan, GPU-driven rendering, PBR, and shader development. Built a 3D Gaussian Splatting renderer and shipped VR/AR experiences on Meta Quest 3 hardware. Skilled in building scalable systems, debugging performance bottlenecks, and collaborating with cross-functional teams to deliver high-quality interactive experiences.

Experience

Indus Trust

March 2025 – Present

AR VR Developer & Engineer

Bangalore

- Built a 3D Gaussian Splatting renderer using C++ and custom HLSL shaders.
- Developed educational XR games using Unreal Engine 5, C++ and HLSL for Meta Quest 3 and 3S.
- Profiled CPU and GPU performance and optimized the game.
- Conducted design and code reviews, and worked in an Agile methodology.
- Participated in meetings and pitched innovative ideas to board members.

Microfacet.io (Volunteering, Non-profit)

April 2025 – December 2025

C++ Developer

Bangalore

- Architected and developed a cross-platform high-performance Physically Based Rendering (PBR) software using Vulkan, modern C++, Slang, GLSL, CMake and RenderDoc.
- Documented and exposed the graphics engine as a public API.
- Attended stand-ups and conducted design and code reviews.

Norian Games

April 2024 – August 2024

Unreal Engine Developer

Remote

- Created a pipeline where the artists and designers can easily collaborate.
- Developed a well-documented custom AI system for NPCs using C++ and integrated it with Unreal Engine 5.
- Debugged and optimized the code to fix the performance issues in the code base.
- Deployed dedicated server builds on a Linux server for multiplayer.
- Conducted research and participated in design sprints using Agile methodologies.

Freelancing

September 2023 – April 2024

Game Developer

Newcastle upon Tyne, UK

- Developed and optimized custom C++ game modules for clients using Unreal Engine.
- Created efficient AI behavior systems and integrated them into existing codebases.
- Developed a RESTful API in C++ to facilitate seamless communication with an HTTP server.
- Delivered a game for a private client [Demo Link](#)

StuBrew - Newcastle University

May 2023 – August 2023

Game Developer

Newcastle upon Tyne, UK

- Designed and developed a Virtual Reality (VR) application using Unreal Engine 5, C++ and blueprint for HTC Vive Pro 2 and Oculus Rift S hardware. [Demo Link](#)
- Programmed the VR interactions with the environment, shaders and an Objective system to tell the user what to do.
- Collaborated with local student brewery from university to design and develop the interactive learning experience they needed.

Education

Newcastle University

Sep. 2022 – Sep 2023

Master of Science in Computer Game Engineering (Distinction)

Newcastle, UK

Sri Manakula Vinayagar Engineering College

Sep. 2017 – Oct 2021

Bachelor of Technology in Computer Science and Engineering (Merit)

Puducherry, India

Projects

Agni Game Engine | *C++, Slang, Vulkan, ImGui, CMake, Git, Github, Visual Studio*

January 2025

- Developed a Data-driven game engine with a custom Vulkan 1.4 renderer and open source tools and libraries for physics and UI.
- Implemented Physically-Based Rendering (PBR) using a metallic-roughness workflow.
- Integrated Directional, Spot, and Point lights with shadow mapping, skybox support, and MSAA (Multi-Sample Anti-Aliasing).
- Implemented full GPU driven rendering by vertex pulling, bindless resources, GPU culling and Indirect draw calls.
- Utilized Dear ImGui for on-screen UI rendering and Jolt Physics for collision and simulation.
- Source code can be found [here](#)

Procedural Grass Rendering | *HLSL, C#, Unity, Git, Github, Visual Studio*

October 2025

- Developed a GPU-driven procedural grass rendering system capable of drawing over one million animated blades of grass in a single draw call using **Compute Shaders**.
- Implemented GPU-based culling and LOD techniques for high performance, achieving over 500 FPS at 1440p on desktop GPUs.
- Designed a noise-driven wind simulation for realistic grass movement with minimal CPU overhead.
- Optimized the rendering pipeline for VR platforms (Meta Quest 3/3S), maintaining 72+ FPS with half a million blades rendered.
- Source code can be found [here](#)

PaintGame | *OpenGL, C++, CMake, Visual Studio, GLSL, Custom game engine, Git, Github* **February 2023 - May 2023**

- Worked as part of a team of eight and made a custom game engine using OpenGL and C++.
- Implemented the renderer for PC and PS4, particle system using instance rendering and gameplay features.
- Designed and executed a multi-threading system to efficiently load resources, improving performance.
- Achieved a grade of 77% for this project.
- By working in a team, I understood how important it is to communicate with people and how much you can achieve when you work together.
- [Project Source code](#)

War-Psychic | *Unreal Engine 5, C++, Blueprint, Perforce, Linux, Visual Studio*

February 2024 - April 2024

- Programmed the NPC AI and animations using C++, Unreal engine's blueprints and blackboard system.
- Developed the Gameplay Interactions, Game loop, User Interface with proper Main menu, Option menu and player HUD.
- Deployed a cloud server to use a repository to share and store the project on Perforce.
- Designed and planned the project, while making it I realized how important time is and learned how one should scope the project.
- [Find the game at itch.io](#) and [Video Link](#)

Technical Skills

Programming & Scripting: C, C++, C#, CMake, HLSL, Slang, Java, Python, HTML, CSS

Developer Tools & Software: Visual Studio, RenderDoc, Nvidia Nsight, Jira, Git, Github, Perforce

Engines & Frameworks & Libraries: Unity, Unreal, Godot, OpenGL, Vulkan, Dear ImGui, C++ STL